

1. **For separation of Ag-specific from nonspecific human B cells, the following positive selection method is effective?**
 - a. ELISA using specific Abs
 - b. Magnetic beads coated with Abs against CD19
 - c. Ficoll-hypaque separation method
 - d. Ammonium sulfate precipitation method
2. **The following characterize HAT media?**
 - a. T stands for thymidine as an inhibitor of the salvage DNA synthesis pathway
 - b. None of the choices is correct
 - c. A stands for Aminopterin as a substrate for synthetic enzymes
 - d. T stands for thymol that inhibits both pathways of DNA synthesis
 - e. H stands for Hypoxanthine as an inhibitor of DNA synthesis
3. **The following does not characterize polyethylene glycol (PEG) used in hybridoma production?**
 - a. Concentration 40-50%
 - b. High cytotoxicity
 - c. High fusion rate
 - d. Molecular weight 1500-4000
4. **The continued storage of cells at all stages of Mab production is essential because?**
 - a. Throughout cloning, there is a high probability of chromosome loss and consequent loss of Ig secretion
 - b. The risk of infection
 - c. There is always a risk of overgrowth by Ig-nonproducing hybridomas
 - d. All choices are correct
5. **The following method is not used in the purification of Mabs?**
 - a. Protein A affinity chromatography
 - b. Gel filtration according to size
 - c. Ammonium sulfate precipitation
 - d. SDS-polyacrylamide gel electrophoresis
6. **The following is correct about feeder cells used in hybridoma cultures?**

- a. It is necessary to use the same species (Balb/c) as the hybridoma parents
 - b. Peritoneal macrophages are usually used
 - c. Are liver cells that are rich in macrophages
 - d. Are lymph nodes taken from young adult animals
7. **It is inadvisable to maintain myeloma cells in culture indefinitely since?**
- a. They become mortal
 - b. They lose Ab secretion
 - c. They might revert to HGPRT+
 - d. They lose the proliferation property
8. **To test the effectiveness of HAT, myeloma cells are cultured in HAT media. The following results should be obtained?**
- a. Myeloma cells die
 - b. Myeloma cells survive
9. **The following is true about the selection of hybridoma after fusion in HAT medium?**
- a. Unfused B cells die within 10-14 days
 - b. The Ag-specific and nonspecific hybridoma cells survive
 - c. Fused myeloma-myeloma and B-B cells die within 10-14 days
 - d. Unfused myeloma cells die within 10-14 days
10. **In hollow fiber bioreactor during the production of Mabs?**
- a. The extra capillary tubes are filled with sterile media
 - b. If the expected levels of Mab in culture are diminishing, the hybridoma cells should be modified
 - c. The fidelity of the specific Mab production should be monitored regularly due to our ability to control the fluid flow
 - d. The capillary tubes are filled with the hybridoma cells

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
B	A	D	D	D	B	C	A	D	D

11. Human myelomas are difficult to establish in tissue culture because all of the following EXCEPT?

- a. Doubling time in culture is about 70 hours
- b. Are large size
- c. Have abundant RER and mitochondria with high Ig production
- d. Do not carry EBV

12. To test whether the separated hybridoma clones are identical or totally different, the following test could be performed?

- a. Specific agglutination test
- b. Specific immunoelectrophoresis
- c. Specific double immunodiffusion
- d. Competitive inhibition radioimmunoassay

13. The mouse spleen is chosen as a source of B lymphocytes because of the following?

- a. Rich in several immune cells including monocytes and macrophages
- b. Has connective tissue that makes cell release easy
- c. Rich in B cells
- d. Has large size

14. In phage display method for production of human Mab, the following step is not included?

- a. Isolation of mRNA then converting it into cDNA by PCR to amplify the Ig variable region
- b. Isolation of myeloma cells from the blood of the human body
- c. The Ig genes are inserted into phage directly
- d. A gene library of this phage-displayed Ig can be made and selection of the phage that produced the Mab

15. Single cell suspension from mouse spleen could be prepared by?

- a. Specific antibodies are added for cell disruption
- b. Treatment with trypsin for cell separation
- c. Cut spleen into 3-4 pieces followed by washing with media to release cells
- d. Using a mesh strainer after the spleen is minced

16. Selection of HGPRT– myeloma cells is carried out in the presence of?

- a. Cytidine
- b. Thymidine
- c. Formaldehyde
- d. 6-thioguanine

17. The following applies to the fusion protocol between B and myeloma cells?

- a. Cells are incubated in HAT medium for 10-14 days
- b. 5% FCS media is recommended
- c. After addition of PEG, the cells should be incubated in HAT media immediately
- d. PEG is added once in one shot very quickly to avoid toxicity

18. Fetal calf serum used in tissue culture to support the growth of B cells?

- a. Is rich in IgG to prevent contamination
- b. Is IgG depleted by using protein G affinity chromatography

19. Balb/C myeloma cells that stick to flask bottom can be resuspended easily by gentle scraping with a plastic Pasteur pipette without trypsin?

- a. False
- b. True

20. Expansion of hybridoma cells prior to cloning is performed by?

- a. 96-well plates used instead of 8-well plates
- b. Absence of FCS or feeder layer cells
- c. HT medium is replaced by HAT
- d. HAT medium is replaced by HT medium

Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
D	D	C	B	D	D	D	B	B	D

21. The following is not correct for long storage of myeloma cells?

- a. The storage media contains formalin in RPMI for sterility

- b. Using cryotubes
- c. Stored in liquid nitrogen
- d. Cells should be placed first in a polystyrene box at -80°C

22. By EBV transformation, the following human Mabs class could be produced from any individual?

- a. IgG only
- b. IgA only
- c. Any Ig class
- d. IgM only

23. To prove specificity of a Mab?

- a. It must have a high titer against the specific Ag
- b. It should be tested whether it reacts with related Ags or not
- c. It should be immunologically different from other Mabs
- d. It should react with strong affinity with the specific Ag

24. Normal cells usually divide only a limited number of times before losing their ability to proliferate, which is a genetically determined event known as senescence. These cell lines are called?

- a. Finite
- b. Consistent
- c. Continuous
- d. Transformed

25. The following applies for affinity determination of a Mab by inhibition radioimmunoassay?

- a. Serial dilutions of unlabeled Abs are dispensed into assay tubes
- b. Scatchard plot is applied for affinity measurement
- c. Constant amounts of unlabeled Abs are added to each tube
- d. Constant amount of Ag is added to all tubes

26. Thymus cells as feeder layer during hybridoma culturing?

- a. Provide growth factors
- b. Should be obtained from young animals
- c. It is not necessary to use the same species (Balb/c) as the hybridoma parents

- d. All choices are correct

27. Prior to fusion between B cells and myeloma cells, it is essential that?

- a. Cell viability should be more than 50%
- b. Cell density should be not more than 10×10^6 cells/ml
- c. B cells should be pure
- d. The myeloma cells should be in the log phase

28. After the fusion experiment, the purpose of the cloning method is?

- a. Isolation of Ag-specific Mabs
- b. Isolation of single Ag-specific B cells
- c. Isolation of Ag-specific myeloma cells
- d. Isolation of Ag-specific hybridoma cells

29. Isolated immune cells from mouse peritoneum include?

- a. Dendritic cells and mast cells
- b. Immature neurons
- c. Epithelial cells
- d. Macrophages and B cells

30. The following solvents or detergents are not used in the viral inactivation of Mabs?

- a. TNPB and sodium cholate
- b. Caprylic acid
- c. Ammonium sulfate
- d. Triton-100

Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30
A	C	B	A	B	D	D	D	D	C

31. In hollow fiber bioreactor during production of Mabs?

- a. If the expected levels of Mab in culture are diminishing, the hybridoma cells should be killed
- b. The extra capillary tubes filled with sterile media
- c. The fidelity of the specific Mab production should be monitored regularly due to outgrowth of non-producing revert cells
- d. the capillary tubes are filled with the hybridoma cells

32. Isotype determination of a monoclonal Ab means?

- a. Determination of its class and subclass
- b. Determination of its binding strength
- c. Determination of its specificity
- d. Determination of its titer

33. According to density gradient centrifugation of human blood using ficoll-hypaque, the buffy coat contains following cells?

- a. monocytes and RBCs
- b. granulocytes and lymphocytes
- c. lymphocytes and platelets
- d. Plasma and ficoll

34. Thymus cells as feeder layer during hybridoma culturing?

- a. provide growth factors
- b. should be obtained from young animal
- c. All choices are correct
- d. It is not necessary to use the same species (Balb/c) as the hybridoma parent.

35. The following characterize HAT media used in Mab production?

- a. T stands for thymidine as inhibitor of salvage DNA synthesis pathway
- b. H stands for Hypoxanthin as inhibitor of DNA salvage synthesis
- c. A stands for Aminopterin as inhibitor of De novo DNA synthesis
- d. T stands for thymol that inhibit both pathways of DNA synthesis

36. The following is not correct for long storage of myeloma cells?

- a. using cryotubes

- b. the storage media contains formalin in RPMI for sterility.
- c. Cells should be placed first in polystyrene box at -80 OC
- d. Stored in liquid nitrogen

37. If you are provided with 2 monoclonal antibodies for titer determination, Mab1 gave positive results at 1:3000 dilution and negative results in ELISA at 1:6,000 dilution. Mab2 gave positive results at 1:1000 dilution and negative results at 1:2000 dilution.

- A. Which Mab has higher titer? (2 marks)
- b. What is the definition of Mab titer 1 mark

38. Myeloma cells are prone to spontaneous mutation in culture. How could you solve this problem? (answer : Store cells in liquid nitrogen)

39. Suppose you have successfully prepared two hybridoma cells producing two monoclonals x and y. How could you prove that both monoclonals recognize the same or different epitope. Mention the main concept (answer : Competitive binding assay)

40. The following does not apply for propagation of monochromatic radiation.

- a. The mice must be adult
- b. Ascitic fluid produced is rich in the monoclonal antibodies.
- c. prior to hybridoma cell injection, the mice must be injected with pristine
- d. the injected mice can be of any strain other than Balb/c.

Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q40
C	A	C	C	C	B		D